A basket of new fruit varieties is coming your way

By The Economist, adapted by Newsela staff on 05.11.26
Word Count **573**
Level **860L**

Humans have bred crops for desirable traits for many thousands of years. Now, they are looking to gene editing as a way of creating new fruit varieties. Photo credit: billysfam / Shutterstock Photo credit: billysfam / Shutterstock

"You don't notice the seeds in a blackberry until you've tried a seedless one," says Tom Adams. He is the head of Pairwise. It is company in North Carolina. It is working on the first blackberries that don't have seeds.

Pairwise is using gene editing to make its seedless blackberries. Gene editing is a method that lets scientists change the DNA of plants. Genes are pieces of DNA. They contain the instructions for building a living thing. Editing genes can change how a living thing looks. For example, it can change how seeds develop. Technically, these blackberries still have seeds. They are just so small and soft that you don't notice them.

Changing Fruits

Fruits have changed a lot. About 12,000 years ago, humans stopped being hunter-gathers. They stopped traveling to find food. Instead they started to grow their own food. Over thousands of years, humans have grown fruit to their tastes. They picked the plants that bore the tastiest fruits or vegetables and bred them to create new plants. This is called selective breeding. It is also known as artificial selection. This is in contrast to natural selection. In nature, living things that are best suited to their environment survive and have more offspring. These are not always the ones with the tastiest fruits.

But thousands of years of selective breeding have led to tastier, hardier, and often larger fruits and vegetables. Today's peaches are much larger than their wild ancestors, for example. Selectively bred fruits also tend to be sweeter.

Now gene editing allows scientists to make changes to plants even more quickly and precisely. CRISPR is the most popular such technique at the moment. It is a technology that allows scientists to change the DNA of an organism. It can change or remove specific genes. That would be hard to do through selective breeding.

Computer software like artificial intelligence (AI) is also helping scientists design fruit. With AI, computers and machines can learn and solve problems using information. Using AI, scientists are able to discover more about certain traits, such as the chemicals that generate flavor.

Slowing Reactions

Chemical reactions are when substances change into new substances. Scientists are looking to slow chemical reactions in other fruits and veggies to make them look more appealing and even last longer.

For example, avocados (yes, they are a fruit!) turn brown when you cut into them. That's because an enzyme called polyphenol oxidase reacts with the oxygen in the air in a process that causes browning. Enzymes can speed up chemical reactions and are also responsible for the ripening process, which means enzymes can change the texture, color, and flavor of food, too.

GreenVenus, a Californian firm, is using CRISPR to develop avocados that don't turn brown. They changed how the enzyme is made in the fruit. Scientists have also developed mushrooms and potatoes that inhibit the same enzyme. So, slowing that reaction down makes mushrooms and potatoes last longer, and they don't turn brown as fast.

Ready To Eat?

So far, a few fruits that have been gene-edited with CRISPR have gone on sale. That's because it takes a long time to grow most fruit. It can take several years for apple or peach trees to begin bearing fruit.

But as more fruity creations go on sale, companies believe that more people will eat fruit. Pairwise estimates that their seedless blackberries could become as popular as easy-peel mandarins.

Write Prompt

Choose a process or event from the text.

What are the most important details about the process or event?

Quiz

1 Which statement summarizes the section “Ready To Eat?”?

(A) CRISPR has not yet been used to edit genes of any fruits on sale in stores.

(B) Seedless blackberries are common, but they are not as popular as easy-peel mandarins.

(C) Most fruit takes a long time to grow, and both apple and peach trees can take several years to begin bearing fruit.

(D) Only a few fruits with edited genes are on sale, but companies believe when more of these fruits go on sale people will eat more fruit.

2 Which sentence from the article helps the reader understand how people have grown fruit to their tastes?

(A) About 12,000 years ago, humans stopped being hunter-gathers.

(B) They picked the plants that bore the tastiest fruits or vegetables and bred them to create new plants.

(C) Chemical reactions are when substances change into new substances.

(D) That’s because an enzyme called polyphenol oxidase reacts with the oxygen in the air in a process that causes browning.

3 Read the paragraph from the section “Changing Fruits.”

> *Computer software like artificial intelligence (AI) is also helping scientists design fruit. With AI, computers and machines can learn and solve problems using information. Using AI, scientists are able to discover more about certain traits, such as the chemicals that generate flavor.*

What conclusion can the reader make based on the paragraph?

(A) AI can teach scientists new things about fruits.

(B) Scientists think designing fruit is too complex for AI.

(C) AI can only design fruits that have certain chemical traits.

(D) Computers and machines have trouble understanding chemicals.

4 What is the main idea of the article?

(A) Avocados turn brown if they are cut in two.

(B) Scientists are making new fruits using gene editing.

(C) Some blackberries have seeds, but others are seedless.

(D) People stopped traveling to find food about 12,000 years ago.

Answer Key

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